



Samriddha Sirsant

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B.Tech

Mechanical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	7.78 (Current)	2023-Present
Senior Secondary	CGBSE Board	88%	2023
Secondary	CBSE Board	93.6%	2021

EXPERIENCE

• AVISA AUTOMOTIVE

May 2025 - Jul. 2025

Mechanical Engineering intern

Incubated At TIP IIT Hyderabad

- Designed the chassis for a 3-wheeler electric vehicle tailored for vendors, with a total load capacity of 600 kg
- Performed static structural and modal analysis to validate chassis strength, stiffness, and vibrational behavior
- Developed and analyzed an independent suspension system and steering head, optimizing for stability, load-bearing efficiency, and ride comfort

PROJECTS

• AUTONOMOUS ROVER FOR WAREHOUSE RACK INVENTORY MANAGEMENT

Nov 2025-Dec 2025

Inter IIT Tech Meet 14.0

- Designed a modular aluminium-extrusion chassis with a low center of gravity, ensuring structural stability during autonomous navigation, vertical scanning operations up to 1.8 m rack height in constrained environments
- Conceptualized and developed a belt-driven Z-axis scanning mechanism, performing mechanism selection (HoQ), CAD modeling, and stiffness considerations to achieve ± 2 cm vertical positioning accuracy with minimal vibration.
- Designed and validated mechanical subsystems including motor mounts, wheel configuration, encoder gear transmission, camera carriage; supported design decisions through structural analysis and experimental stability testing

• DD-ROBOCON 2026

December 2025 - March 2026

Robotics Club, IIT Indore

- Led a multidisciplinary team approximately 20 members for the development of robots based on the ABU Robocon 2026 problem statement and competition theme, “Kung Fu Bots.”
- Contributed to the mechanical design and overall assembly of multiple robotic mechanisms, including subsystem integration, structural design, and motion-related components required for task execution.
- Successfully qualified the first stage of the competition, securing a score of 90/100 based on design Report, technical evaluation, and animation of the working of different mechanism of the Robots

• DESIGNING ROCKET RE-ENTRY MECHANISM SYSTEM

May 2024 - July 2024

IITI-SOC

- Led a four-member team over a two-month period, securing the Gold Medal at the IITI-SOC competition
- Evaluated multiple rocket steering mechanisms and identified the RCS-parachute combination as the most effective
- Developed CAD models for the parachute and RCS mechanisms and performed analyses to ensure functionality
- Developed Python-based simulation for the rocket's re-entry trajectory, optimizing controlled landing.

• STATIC ANALYSIS OF COMPOSITE STRUTS FOR ANTENNA TRUSS STRUCTURE

Feb 2026 - April 2026

Course Project - Computational Fluids and Structures - Prof Bhargav Vaidya, Prof Sudheer kamble

- Conducted a detailed literature review on antenna truss and struts, and aerospace-grade composite materials, with replication of a published DDTA research paper to understand structural behavior and analysis methodology.
- Performed static structural analysis of the antenna truss assembly by applying boundary conditions given in the research paper to identify critical load-bearing members, evaluate stress distribution.
- Designed and analyzed composite wrapping configurations for the critical strut using ANSYS ACP(Pre) and static structural, comparing different composite materials and ply orientations to achieve improved stiffness, strength, and weight optimization.

• YUKTIYANA - A MULTI TERRAIN ROBOT (UAV + UGV)

August 2024 - Feb 2025

Capstone Project under Prof Dr Vijay A S

- Designed and implemented a transformation mechanism enabling switching between UGV and UAV modes.
- Worked on the Design of the Morphobot maintaining the load distribution, structural stability and correct actuation.
- Analysed the weight distribution and load bearing capacity using static structural analysis in Ansys

- **DESIGN AND DEVELOPMENT OF A 4 LEGGED MOBILE ROBOT (QUADRUPED)** *Feb 2026 - May 2026*
Machine Design Course Project, IIT Indore
 - Designed and developed an 8-DOF quadruped robot actuated using 8 servo motors with a 4-link mechanism.
 - Performed static structural and fatigue analysis on all links to evaluate stress distribution, identify critical regions.
 - Applied machine design principles and failure theories for detailed static and fatigue stress calculations, including stress calculation at the notches and finding the critical location. Also Performed detailed analysis of 4 bar linkage
 - Fabricated the complete robot using PLA-based 3-D printed components and assembled the structure using mechanical fasteners to ensure structural strength and rigidity.

- **NAMASTE BOT** *October 2024 - March 2025*
Robotics Club IIT Indore
 - Designed and assembled a 6-DOF mechanical robotic hand made up of PLA as a material and servo motor for actuation of joints capable of performing the “Namaste” gesture and other basic movements
 - Oversaw the complete manufacturing process, including 3D printing of all components, and led the assembly.
 - Worked on kinematics and the mechanical joint configurations

GO-KART *August 2024 - August 2025*

IVDC CLUB IIT INDORE

- Making of A cart from scratch with complete cad design of all the parts, manufacturing and assembling
- Designed and did thermal analysis of the model of break pedal assembly
- Worked on the manufacturing of Chassis using Cutter, Grinder and Welding of the rods

TECHNICAL SKILLS

- **CAD DESIGN AND ANALYSIS:** Fusion 360 , Solidworks , Ansys, Autocad, Abaqus
- **Programming:** Python*, C++*, MATLAB, Simulink* , Simscape* ** Elementary proficiency*
- **Video Editing:** Da Vinci Resolve
- **Others:** 3D - printing, operating Laser cutter, LabView*
- **Soft Skills:** Communication, Team Work, Presentation, Management, Problem-Solving, Project Planning, Leadership

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Basic Calculus, Differential equation, Numerical Methods
- **Core Mechanical:** Solid Mechanics, Fluid Mechanics, Thermodynamics, Principles of Industrial Engineering, Design Thinking for Mechanical Engineers, Material Science and Engineering, Fundamentals of Vibrations, Theory of Manufacturing Processes, Introduction to Additive Manufacturing, Fluid Machinery, Strength of Materials, Fundamentals of Acoustics, Water Resources Planning and Management. Heat Transfer, Instrumentation and Control System, Machining Science and Metrology, Introduction to Robotics, Introduction to Finite Element Methods, Trends and Development in Solid Mechanics, Introduction to Hybrid Additive Manufacturing Techniques, Machine Design, Kinematics and dynamics of machine, Applied Thermodynamics, Computational Fluid Dynamics, Automobile, Electric Vehicles.
- **Lab Course:** Solid Mechanics Lab, Fluid Mechanics and Machinery Lab, Manufacturing Processes Lab, Makerspace, Computer Aided Machine Drawing, Heat transfer Lab, Machining Science and Metrology Lab, Instrumentation and Control System Lab.

POSITIONS OF RESPONSIBILITY

- **Co-Head,** Robotics Club, IIT Indore *March 2025 - Present*
- **Video Editing Head,** VLR club, IIT Indore *Aug 2024 - March 2025*
- **Member,** CAE club, IIT Indore *May 2024 - Present*
- **Member,** IVDC club, IIT Indore *August 2024 - March 2025*
- **Mentor,** IITI-Summer of codes, IIT Indore *May 2025 - Aug 2025*
- **Mentor,** Tech-eshish Castle, Robotics Club *Jan 2025 - Feb 2025*

ACHIEVEMENTS

- **BRONZE MEDAL;** Inter IIT Tech Meet 14.0 (Associate) *2025*
- **GOLD MEDAL,** IITI-SOC design domain among approximately 100 teams *2024*
- **JEE ADVANCED,** Cleared JEE advanced and was among top 1 percent among country *2023*
- **NTSE,** cleared NTSE Stage-1 *2020*

INCLUDE ALONGSIDE PROJECTS

- **DRIVE** , https://drive.google.com/drive/folders/1K1CnxRvxaq3rV5-2I2DD6lwgg65R8f3f?usp=drive_link *2024*